THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — SCIENCE · HARD LEVEL CHAPTER TEST SET 1

Chapter 5: **Life Processes** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Assertion (A): The rate of breathing in aquatic organisms is much faster than in terrestrial organisms. Reason (R): The amount of dissolved oxygen in water is fairly low compared to air. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. In aerobic respiration, one molecule of glucose gives how many molecules of carbon dioxide? 1

(a) 2 **(b)** 6 **(c)** 1 **(d)** 12

3. A closed jar contains a green plant in sunlight together with a mouse. The mouse survives longer than a mouse alone in a jar because the plant: 1

(a) uses carbon dioxide and releases oxygen **(b)** absorbs oxygen only
(c) gives out nitrogen **(d)** releases carbon dioxide

4. Assertion (A): Veins have valves but arteries do not. Reason (R): Blood in veins flows at low pressure and must be prevented from flowing backwards. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

5. In photosynthesis, six molecules of carbon dioxide combine with water to form how many molecules of glucose? 1

(a) 1 **(b)** 12 **(c)** 2 **(d)** 6

6. A leaf variegated with green and white patches shows starch only in green regions because: 1

(a) white parts get no light **(b)** starch is stored there
(c) chlorophyll is needed for photosynthesis **(d)** white parts are dead

7. Assertion (A): Bile juice is necessary for digestion of fats. Reason (R): Bile contains the enzyme lipase that digests fats. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

8. A heart beats 72 times per minute. The number of beats in 1 hour is: 1

(a) 432 **(b)** 720 **(c)** 4320 **(d)** 7200

9. **Assertion (A):** Guard cells control the opening and closing of stomata. 1
Reason (R): Guard cells swell when water flows into them, causing the stomatal pore to open.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** Arteries have thick, elastic walls. 1
Reason (R): Veins have valves that prevent backflow of blood.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Calculate the volume of blood pumped in 1 minute if the heart beats 72 times per minute and pumps 70 mL each time. 2

12. Why is the pH of the small intestine alkaline and why is that necessary? 2

13. If a person has anaemia, why does he feel tired? 2

14. The glomerular filtrate is 180 L per day and urine is 1.5 L. Find the volume of water reabsorbed and the percentage. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. In an experiment, the destarched leaf was covered with black paper partly. After sunlight and iodine test the uncovered part turned blue-black. Interpret fully. 3

16. Explain why the human heart muscle needs its own blood supply (coronary arteries). 3

17. Describe how digestion of starch occurs from the mouth to the small intestine. 3

OR

Explain why a person who is climbing high mountains breathes faster and has a higher RBC count. 3

18. A sample of urine shows glucose. Which part of the nephron is faulty? Explain the normal process. 3

OR

Explain the absorption of water by the roots and the role of osmosis. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Meera had a lunch of chapatti, dal and a glass of milk. The food first entered her mouth, where it was mixed with saliva, and then passed through the oesophagus into the stomach and later into the small intestine. 5
(i) Name the enzyme in saliva that acts on chapatti.
(ii) Name the acid secreted in the stomach.
(iii) Where does the complete digestion of food take place?
(iv) State the role of bile juice in the digestion of fat in milk and explain why the medium in the small intestine must be alkaline.

20. Calculate: a person breathes 15 times per minute with 500 mL per breath. Find the volume of air in an hour and the oxygen inhaled if air has 21% oxygen. 5

OR

An athlete's blood pressure is measured as 120/80 mm Hg. Explain the meaning of the terms, the instrument used and the effect of hypertension. 5

END OF QUESTION PAPER — Total: 40 Marks